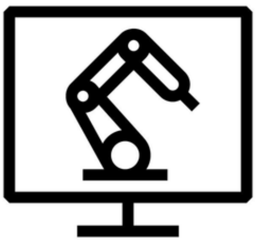
Robotics Services

Two services for robotics teams:

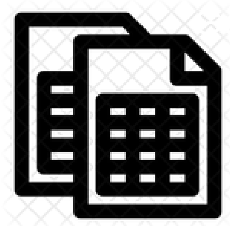
- A **clean simulation world** for your robot
- Labelled **synthetic data** for your models

What We Offer



Simulation Setup

Robot model, workspace, sensors, parts, lighting, and physics - built for your robot.



Synthetic Data

Labelled images, masks, depth, pose, sensor data, and demonstrations.

BUILT WITH

Gazebo · Isaac Sim

SHIPPED IN

LeRobot · COCO · YOLO · HDF5 · MCAP

Learn more at *dodao.io*/robotics

Simulation Setup

When you start a robotics project, the first step is building a **clean simulation world**. We build that world for you so your team can focus on the parts that are unique to your product.

What We Build

Every simulation world has six pieces. We build all six for your specific use case, then ship the project folder.

Robot Model

The robot with the correct joint chain, link frames, and joint limits. Loads cleanly in ROS 2 and MoveIt 2.

Workspace

The bench, table, floor, fixtures, and safety enclosures around the cell. Matches the real dimensions you give.

Parts

The objects the robot picks, places, or scans at the right size, weight, and material for realistic grips.

Sensors and cameras

Cameras, depth sensors, and force/torque sensors, placed where they sit on the real cell for in sim testing.

Lighting and materials

Light sources and surface materials matched to the real environment so vision models trained in sim work on camera.

Physics and contacts

Friction, mass, and joint dynamics tuned so grasping, insertion, and stacking behave the same in sim as on hardware.

Why Start Here

- Catch problems early. Spot reach, sensor, or layout issues before any hardware is built.
- Speed: Done in weeks. A working scene ready in weeks, not months.
- Right tools picked for you. We pick Gazebo or Isaac Sim based on your robot.

Tools We Use

Gazebo

ROS 2 native. Free.
Quick to iterate.

Learn more

Isaac Sim

Realistic images.
Built for AI training

Learn more

Synthetic Data

Real-world data is **slow and expensive** to label. Synthetic data is rendered inside the simulator, where the system already knows the ground truth of every pose, every contact force, and every pixel.

What We Produce

Detection images and masks

Camera frames with bounding boxes plus segmentation masks at every pixel. Ready to train YOLO or any model your team already runs.

Depth, pose, and grasp labels

Depth at every pixel plus the position, orientation, and grasp points of every object. The data picking and assembly models need.

Demonstration trajectories

Full observation and action logs of a teleoperated expert doing the task. The training set for behaviour cloning, diffusion policy, and VLA models.

Rare and edge cases

Occlusions, damaged goods, glare, spills, and people in the path. Generated on purpose so your model sees what real data rarely captures.

Non camera sensors

Lidar point clouds, thermal frames, ultrasonic returns, and force readings with realistic noise. For robots that rely on more than cameras.

OCR/barcode renders

Camera frames with generated text, codes, and labels at random fonts, glare, and angles. The training data OCR and barcode pipelines need.

How We Generate It

Domain Randomization | Procedural Scenes | Photo Real Rendering | Sensor Noise Modelling

Shipped In Your Format

LeRobot | Robomimic | RLDS | COCO JSON | Custom HDF5 | MCAP rosbags | YOLO labels

Tell us about your project

We help robotics teams build the two pieces of infrastructure that come before training and testing: a **clean simulation world** for your robot, and the **labelled synthetic data** your models need to learn from.

We will reply within two business days with a written proposal which includes scope, timeline and cost.

Connect With Us